ALMA MATER STUDIORUM
UNIVERSITÀ DI BOLOGNA

DEPARTMENT OF PHYSICS AND ASTRONOMY "A. RIGHI"

SECOND CYCLE DEGREE

PHYSICS

Critiquing Pure Reason in proximity of a Hollow-Core Fiber

Supervisor

Prof. Francesco Minardi

Co-supervisor optional

Prof. Qualcuno

Defended by

Francesco Nasoni

Graduation Session July 2026

Academic Year 2025/2026

Pokémon forti, Pokémon deboli.
Sono distinzioni dettate dall'egoismo.
Gli allenatori davvero in gamba dovrebbero vincere coi loro preferiti.

Strong Pokémon, weak Pokémon.
That is only the selfish perception of people.
Truly skilled trainers should try to win with their favorites.

Karen, Johto Elite Four

Contents

Introduction

Your introduction text here.

Chapter 1

Theoretical Background

Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetuer id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

1.1 Optical Dipole Trap

Optical traps are experimental tools that allow the confinement of atom or ions within a localized region of space, under the condition that these particles have a kinetic energy lower than the trap potential depth (typically below 1mK). For the work presented in this thesis, among all possible trapping techniques, the focus will be strictly placed on Optical Dipole Traps (ODTs) for neutral atoms.

Optical dipole traps rely on the electric dipole interaction of the atoms with far-detuned light. Following the work of Grimm et al. [1], this section reviews the theory behind the working mechanism of this technique, particularly in the case of red-detuned ODTs.

1.1.1 Oscillator model

The optical dipole force arises from the interaction of the induced atomic momentum with the intensity gradient of the light field. It is a conservative force, thus in the following its potential energy will be derived modelling the atom as an oscillator subjected to a classical oscillating electric field.

1.1.1.1 Interaction between induced dipole moment and driving field

The laser field is modeled as the oscillating electric field:

$$\mathbf{E}(\mathbf{r},t) = \hat{\mathbf{e}}\mathcal{E}(\mathbf{r})e^{-i\omega t} + c.c.\ , \tag{1.1}$$

where $\hat{\mathbf{e}}$ is the unitary polarization vector and $\mathcal{E}(r)$ is the complex amplitude of the field. Denoting the atomic polarizability as α, the induced dipole momentum of the atom is given by:

$$\mathbf{p}(\mathbf{r},t) = \alpha\mathbf{E}(\mathbf{r},t) = \hat{\mathbf{e}}\alpha\mathcal{E}(\mathbf{r})e^{-i\omega t} + c.c. \tag{1.2}$$

In the semiclassical framework, the interaction of a neutral atom with a laser light field can be described using the dipole approximation. The interaction Hamiltonian takes the form $H_{\text{dip}} = -\mathbf{p}\cdot\mathbf{E}$. For an induced dipole, where the dipole moment depends linearly on the external field, the potential energy is obtained by integrating the work required to induce the dipole as the electric field is increased from zero to its final value:

$$U_{\text{dip}}(t) = \int_0^{\mathbf{E}} -\alpha\mathbf{E}'\cdot d\mathbf{E}' = -\frac{1}{2}\alpha E^2 = -\frac{1}{2}\mathbf{p}\cdot\mathbf{E}\,. \tag{1.3}$$

Averaging over a field period [1]:

$$U_{\text{dip}} = -\frac{1}{2}\langle\mathbf{p}\cdot\mathbf{E}\rangle = -\frac{1}{2\epsilon_0 c}\text{Re}[\alpha]I, \tag{1.4}$$

where $I = 2\epsilon_0 c|\mathcal{E}|^2$ is the time-averaged field intensity.
This result shows that an atom in an oscillating laser field experiences a force proportional to the field intensity I, and to the real part of the polarizability, which represent the in-phase component of the dipole oscillation, responsible for the dispersive properties of the interaction.
It follows that the dipole force is a conservative force proportional to the electric field intensity gradient:

$$\mathbf{F}_{\text{dip}}(\mathbf{r}) = -\nabla U_{\text{dip}} = \frac{1}{2\epsilon_0 c}\text{Re}[\alpha]\nabla I(\mathbf{r}). \tag{1.5}$$

The dipole oscillation absorbs a power

$$P_{\text{abs}} = \langle\dot{\mathbf{p}}\mathbf{E}\rangle = 2\omega\text{Im}[\alpha\mathcal{E}^2] = \frac{\omega}{2\epsilon_0 c}\text{Im}[\alpha]I, \tag{1.6}$$

which is proportional to the out-of-phase component of the oscillating dipole. The absorbed power can be seen as series of absorption and spontaneous emission of photons with energy $\hbar\omega$. Therefore, the total scattering rate is given by

$$\Gamma_{\text{sc}}(\mathbf{r}) = \frac{P_{\text{abs}}}{\hbar\omega} = \frac{1}{\hbar\epsilon_0 c}\text{Im}[\alpha]I(\mathbf{r}). \tag{1.7}$$

1.1.1.2 Atomic polarizability

The polarizability α can be calculated by considering the classical Lorentz oscillator model for the atom, according to which the polarizability can be calculated by solving the differential equation

$$\ddot{x} + \Gamma(\omega)\dot{x} + \omega_0^2 x = -\frac{eE(t)}{m_e}, \tag{1.8}$$

Using $E(t) = \mathcal{E}_0 \exp(-i\omega t)$, thus looking for a solution in the form $x(t) = x_0 \exp(-i\omega t)$, equation (1.8) gives

$$x_0 = -\frac{e\mathcal{E}_0}{m_e}\frac{1}{\omega_0^2 - \omega^2 - i\omega\Gamma(\omega)}. \tag{1.9}$$

Using $p = -ex(t) = -ex_0 \exp(-i\omega t)$ and (1.9), one can easily obtain the polarizability

$$\alpha(\omega) = \frac{e^2}{m_e}\frac{1}{\omega_0^2 - \omega^2 - i\omega\Gamma(\omega)}, \tag{1.10}$$

where $\Gamma(\omega)$ is the classical damping rate due to the radiative energy loss:

$$\Gamma^{\text{class.}}(\omega) = \frac{e^2\omega^2}{6\pi\epsilon_0 m_e c^3}. \tag{1.11}$$

Defining the on-resonance spontaneous emission rate $\Gamma \equiv \Gamma^{\text{class.}}(\omega_0) = (\omega_0/\omega)^2\,\Gamma^{\text{class.}}(\omega)$, the polarizability becomes

$$\alpha(\omega) = 6\pi\epsilon_0 c^3 \frac{\Gamma/\omega_0^2}{\omega_0^2 - \omega^2 - i(\omega^3/\omega_0^2)\Gamma}. \tag{1.12}$$

When considering a full quantum picture of a two level system subjected to classical radiation, Γ should be substituted with the spontaneous emission rate

$$\Gamma = \frac{\omega_0^3}{3\pi\epsilon_0\hbar c^3}|\langle e|\,\hat{d}\,|g\rangle|^2, \tag{1.13}$$

where $\hat{d}$ is the dipole moment operator.

Nevertheless, optical dipole traps work in far-detuned low saturation regime. Consequently, the transitions to the excited state are strongly suppressed, and the atom can be considered as occupying the ground state only. In this limit, the atomic equation of motion reduces to a classical damped harmonic oscillator model described by equation 1.8 (for a rigorous derivation see Chapter V of Steck [2]). Therefore, the classical polarizability given by equation 1.12 represent a well suited approximation for ODTs.

1.1.1.3 Dipole potential and scattering rate

Substituting α form (1.12) in (1.4) and (1.7) one gets the expressions of the dipole potential and the scattering rate of a dipole trap as explicit function of ω, ω_0 and Γ:

$$U_{\text{dip}}(\mathbf{r}) = -\frac{3\pi c^2}{2\omega_0^3}\left(\frac{\Gamma}{\omega_0-\omega}+\frac{\Gamma}{\omega_0+\omega}\right)I(\mathbf{r}), \tag{1.14}$$

$$\Gamma_{\text{sc}}(\mathbf{r}) = \frac{3\pi c^2}{2\hbar\omega_0^3}\left(\frac{\omega}{\omega_0}\right)^3\left(\frac{\Gamma}{\omega_0-\omega}+\frac{\Gamma}{\omega_0+\omega}\right)^2 I(\mathbf{r}). \tag{1.15}$$

These equations show two resonance frequency for $\omega = \pm\omega_0$. Introducing the detuning $\Delta = \omega - \omega_0$, for the far-red regime it holds that $\Gamma \ll |\Delta|$. For instance, the D_2 line in Rb atoms has a transition frequency of ~ 3.8 THz thus a trap typically works at few Thz of detuning, while $\Gamma \sim 36$ MHz. This allows the introduction of the well-known rotating-wave approximation, according to which the anti-resonant term can be neglected, further simplifying the equations to

$$U_{\text{dip}}(\mathbf{r}) = \frac{3\pi c^2}{2\omega_0^3}\frac{\Gamma}{\Delta}I(\mathbf{r}) \tag{1.16}$$

$$\Gamma_{\text{sc}}(\mathbf{r}) = \frac{3\pi c^2}{2\hbar\omega_0^3}\left(\frac{\Gamma}{\Delta}\right)^2 I(\mathbf{r}) = \frac{\Gamma}{\hbar\Delta}U_{\text{dip}} \tag{1.17}$$

Equations (1.16) and (1.17) are the fundamental equations governing the behavior of optical dipole traps and yield important physical considerations for their implementation. The sign of U_{dip} is determined by the sign of the detuning Δ. For red detuning ($\Delta < 0$) the potential is negative, confining atoms at intensity maxima, whereas for blue detuning ($\Delta > 0$), the potential is positive pushing atoms to intensity minima. Because the potential depth scales as I/δ while the scattering rate scales as I/Δ^2, optical traps are typically realized with high intensity field and large detunings, in this way, the trap depth is maximized while maintaining a low scattering rate.

1.1.2 Multi-level atoms

Moving from the two-level atom to a real atom, the electronic transitions become more complex, and the calculation of the dipole potential must take into consideration all the possible allowed transitions. This lead to a dipole potential that depends on the particular electronic substate.

1.1.2.1 Ground-state light shifts

Labelling as ε_i the energy levels of the multi-level atom, the effect of the laser light on the levels can be described by means of the non-degenerate time-independent second order perturbation theory. Considering the interaction Hamiltonian $\mathcal{H}_I = -\hat{\mathbf{d}} \cdot \mathbf{E}$, the perturbative correction on the i-th energy level is

$$\Delta E_i = \sum_{j \neq i} \frac{|\langle j|\mathcal{H}_I|i\rangle|}{\varepsilon_i - \varepsilon_j}. \tag{1.18}$$

The calculation can be developed by employing the dressed atom approach, in which the quantum states describe the combined atom-field system. Defining the unperturbed atomic ground state energy as zero, in this framework an unperturbed dressed state has an energy $\varepsilon_i = n\hbar\omega$ that depends solely on the number of photons n in the field. The absorption of a photon transitions the system to another dressed state characterized by the energy $\varepsilon_j = \hbar\omega_j + \hbar(n-1)\hbar\omega$, where $\hbar\omega_j$ represent the bare atomic transition energy. In such a way, the denominator of equation (1.18) reduces to

$$\varepsilon_i - \varepsilon_j = -\hbar\omega_j + \hbar\omega = -\hbar\Delta_{ij}, \tag{1.19}$$

where Δ_{ij} is the detuning of the $i \to j$ transition with respect to the laser field.

Considering a two-level atom first, (1.18) has only one term, using equations (1.13) and (1.19) and $I = 2\epsilon_0 c|\mathcal{E}|^2$, one gets:

$$\Delta E = \pm \frac{|\langle e|\mu|g\rangle|^2}{\Delta}|E|^2 = \pm \frac{3\pi c^2}{2\omega_0^3}\frac{\Gamma}{\Delta}I, \tag{1.20}$$

for the ground and excited state (upper and lower sign, respectively). Equation (1.20) shows that the energy shift of the ground state (the solution with the sign +) exactly correspond to the dipole potential (1.16). This allows the reinterpretation of the semi-classical result obtain for the external degrees of freedom (position and momentum) in terms of the dynamic of the internal degrees of freedom. In presence of the laser field, the energy levels are shifted by a quantity proportional to the field intensity with a sign that depends on the sign of the detuning. If a low saturation regime is assumed ($I/\Delta \ll 1$), all the atom can be considered in the ground state, and the light-shifted ground state became the effective potential that govern the motion of the atoms which drives them toward region of minima of ΔE (Figure 1.1).

For a multi-level atom equation (1.18) contains many terms similar to equation (1.20), accounting for all the allowed transitions from any ground state sublevel $|g_i\rangle$ to any excited state $|e_j\rangle$. However, each of these terms depends on the dipole matrix element $d_{ij} = \langle g_i|\hat{d}|e_j\rangle$. Such quantity can be expressed as

$$d_{ij} = c_{ij}||d|| \tag{1.21}$$

where $||d||$ is the reduced matrix element given by (1.13) and c_{ij} are real transition coef-

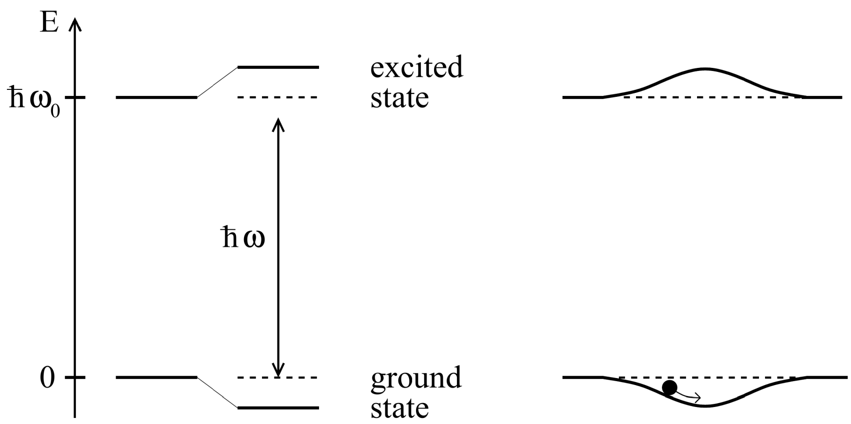

Figure 1.1: *From [1]. Left: light-shifted energy levels for a two-level atom for a red detuning ($\Delta < 0$). Right: a spatially inhomogeneous field produces a potential well that traps the atoms.*

ficients that take into account the coupling strength of the two involved states $|g_i\rangle$ and $|e_j\rangle$, and depends on their angular momentum value [1][3].
Therefore, the energy shift (1.18) for the ground state $|g_i\rangle$ is

$$\Delta E_i = \frac{3\pi c^2\Gamma}{2\omega_0^3} I \times \sum_j \frac{c_{ij}^2}{\Delta_{ij}} \tag{1.22}$$

and according to the interpretation given from the two-level atoms, the dipole potential at low saturation is

$$U_{\text{dip}} = \Delta E_i. \tag{1.23}$$

Thus, the dipole potential in a multi-level atom arises from the contribution of all the allowed transition, each weighted by the ratio between its line strengths c_{ij}^2 and its detuning Δ_{ij}.

1.1.2.2 Fine and Hyperfine structures contribution for Alkali atoms

Many laser cooling experiments utilize alkali metal atoms. This preference arises because their principal transition frequencies fall within the VIS-IR range making them easily accessible with standard laser technologies. Moreover, their relatively simple atomic structure are particularly convenient, allowing easy identification of closed transitions necessary for continuous cooling cycles, while minimizing the number of repumper lasers needed to address the population leaking to dark states.
When considering the fine structure arising from the spin-orbit interaction, the electronic ground state of alkali atoms is ${}^2S_{1/2}$. From this state, the electric dipole selection rules ($\Delta L = \pm 1$, $\Delta S = 0$) allow for two primary transitions, known as $D-$lines:

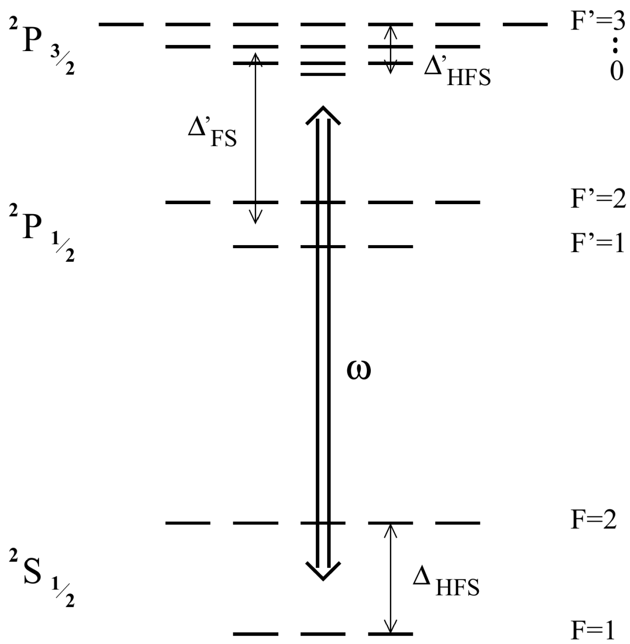

Figure 1.2: *From [1]. Energy level scheme of an alkali atom with* $I = 3/2$ *such as* ^{87}Rb.

- D_1: $^2S_{1/2} \to {}^2P_{1/2}$;
- D_2: $^2S_{1/2} \to {}^2P_{3/2}$.

Beyond the fine structure, the energy levels undergo further splitting due to the coupling between the nuclear spin and the total electronic angular momentum. The resulting configuration is called hyperfine structure. Denoting the fine structure splitting as $\hbar\Delta_{FS}$, the ground state hyperfine splitting as $\hbar\Delta_{HFS}$, and the excited state hyperfine splitting as $\hbar\Delta'_{HFS}$, these characteristic energy scales obey $\Delta_{FS} \gg \Delta_{HFS} \gg \Delta'_{HFS}$. The complete electronic energy level scheme for an alkali atom with nuclear spin $I = 3/2$ is shown in figure 1.2.

Assuming that all optical detunings stay large compared with the excited-state hyperfine splitting Δ'_{HFS}, equation (1.22) can be calculated for an alkali atom in the ground state of total angular momentum F and magnetic quantum number m_F, obtaining [1]:

$$U_{\text{dip}}(\mathbf{r}) = \frac{\pi c^2 \Gamma}{2\omega_0^3} \left(\frac{2 + \mathcal{P} g_F m_F}{\Delta_{2,F}} + \frac{1 - \mathcal{P} g_F m_F}{\Delta_{1,F}} \right) I(\mathbf{r}), \tag{1.24}$$

where the two term in the parentheses represent respectively the contribution of the D_2 and D_1 line. Here, g_F is the Landé factor, $\mathcal{P}$ characterizes the laser polarization ($\mathcal{P} = 0, \pm 1$ for linear polarization and circular $\sigma^\pm$ respectively), and $\Delta_{2,F}$ and $\Delta_{1,F}$ denote the detunings of the laser field relative to the transitions from the $^2S_{1/2}, F$ ground state to the center of respectively the $^2P_{3/2}$ and $^2P_{1/2}$ hyperfine excited state.

If the detunings are large with respect to the fine-structure splitting ($|\Delta_{1,F}|, |\Delta_{2_F}| \gg$

Δ'_{FS}), introducing the detuning Δ with respect to the center of the $D-$line doublet, expanding equation (1.24) at first order of the small parameter Δ'_{FS}/Δ, one gets [1]:

$$U_{\mathrm{dip}}(\mathbf{r}) = \frac{3\pi c^2}{2\omega_0^3}\frac{\Gamma}{\Delta}\left(1 + \frac{1}{3}\mathcal{P} g_F m_F \frac{\Delta'_{\mathrm{FS}}}{\Delta}\right) I(\mathbf{r}). \tag{1.25}$$

The zeroth order term corresponds exactly to the dipole potential of a simple two-level atom (see Eq. (1.16)). The first-order term introduces a small, spin-dependent correction depending on the laser polarization $\mathcal{P}$ and the magnetic quantum number m_F.

Physically, equation (1.25) describes that, when the fine structure is completely unresolved due to the large laser detuning, the atom effectively acts as a generic two-level system undergoing an $s \to p$ transition. Consequently, the resulting dipole potential is dominated by the scalar term, matching the two-level model previously discussed.

Since in the unresolved fine structure case, the polarization-dependent term is heavily suppressed by the small parameter Δ'_{FS}/Δ (and vanishes entirely for linearly polarized light), the atom effectively acquires only the scalar light shift of the unperturbed s state. Consequently, all hyperfine and magnetic sublevels experience a uniform trapping potential that confines the atom while preserving its internal spin degeneracy.

In a more general case, when the fine structure is resolved, but the hyperfine structure is unresolved ($|\Delta_{1,F}|, |\Delta_{2,F}| \gg \Delta_{HFS} \gg \Delta'_{HFS}$), only linearly polarized light guarantees that the dipole potential becomes completely independent of m_F and F [1]. Therefore, linearly polarized light is usually preferred for dipole trap implementation.

1.1.3 Heating mechanism

Typical trap depths for optical dipole traps are usually below 1 mK. Therefore, capturing a substantial number of atoms requires a preliminary cooling stage. For instance, the experiment discussed in this thesis employs a magneto-optical trap (MOT) followed by an optical molasses phase.

Given these shallow confining potentials, even minor heating rates can largely limit the trap lifetime. It is therefore important to outline the mechanisms responsible for the parasitic heating induced by the trapping light.

1.1.3.1 Heating source

When atoms interact with laser light, fluctuations in the spontaneous emission and absorption always causes heating (this is also the phenomenon that puts a lower limit to cooling mechanism [4][3]). For the optical dipole trap the atoms absorbs photons and re-emits them in random directions (isotropic radiation), statistical oscillations in both these processes cause heating. The absorption of photons from the dipole light is associated

with a recoil energy $E_{\rm rec} = k_B T_{\rm rec}/2$ per scattering event and happens only in the direction of the laser beam, hence it is anisotropic. On the other hand, the re-emition scattering events are also associated with a recoil energy $E_{\rm rec}$, however the emission directions are random, thus the effect is spread over all the three spatial direction. Thus, for a single cycle the atom receives $E_{\rm rec}$ from the absorption and then $E_{\rm rec}$ from the spontaneous emission, for a total of $2E_{\rm rec}$, in a time $\bar{\Gamma}_{\rm sc}^{-1}$.
It leads to the total heating power

$$P_{\rm heat} = 2E_{\rm rec}\bar{\Gamma}_{\rm sc} = k_B T_{\rm rec}\bar{\Gamma}_{\rm sc}, \tag{1.26}$$

the heating depends on the mean scattering rate $\bar{\Gamma}_{\rm sc}$ and on the recoil $E_{\rm rec}$ which depends on the laser frequency.
Other fundamental source of heating, such as the redistribution of photons between different travelling wave, should be quenched by the far-detuned working frequency [1].

1.1.3.2 Heating rate

Assuming that the trapped atoms are at thermal equilibrium, the mean energy can be expressed through the energy equipartition theorem. Assuming the trap as a three-dimensional harmonic potential well, there is a contribution of three quadratic degrees of freedom from the kinetic energy, and other three from the harmonic potential. According to the energy partition theorem, the mean energy is

$$\bar{E} = 3k_B T. \tag{1.27}$$

Using equation (1.26) in equation (1.27), one can obtain an equation that describes the temperature change with time:

$$\dot{T} = \frac{1}{3}T_{\rm rec}\bar{\Gamma}_{\rm sc}. \tag{1.28}$$

Expressing the dipole potential as

$$\bar{U}_{\rm dip} = U_0 + \bar{E}_{\rm pot} = U_0 + \frac{3}{2}k_B T, \tag{1.29}$$

where U_0 is the potential energy minimum, then the scattering rate $\bar{\Gamma}_{\rm sc}$ can be expressed by Substituting $U_{\rm dip}$ from (1.17), obtaining

$$\bar{\Gamma}_{\rm sc} = \frac{\Gamma}{\hbar\Delta}(U_0 + \frac{3}{2}k_B T). \tag{1.30}$$

It is important to notice that, for a blue-detuned trap, $U_0 = 0$ and the trap depth $\hat{U}$ is given by the maximum value of the potential surrounding the minimum, by contrast for a red-detuned trap $\hat{U} = |U_0|$. Finally, using equation (1.30) in (1.28), with a depth

$\hat{U} \gg k_B T$, one gets the following temperature changing rate for a red and a blue trap:

$$\dot{T}_{\text{red}} = \frac{1}{3} T_{\text{rec}} \frac{\Gamma}{\hbar|\Delta|} \hat{U}, \tag{1.31a}$$

$$\dot{T}_{\text{blue}} = \frac{1}{2} T_{\text{rec}} \frac{\Gamma}{\hbar\Delta} k_B T. \tag{1.31b}$$

Equations (1.31) describe the increasing in temperature of the atomic cloud if subjected to the dipole trap only. It highlights another blue trap limitation: while a red trap has a linear heating rate (as long as $\hat{U} \gg k_B T$), the blue trap shows an exponential heating law.

1.2 Optical Fibers

1.2.1 Circularly symmetric wave guides

1.2.2 Weak waveguide approximation

1.2.3 The LP modes

Chapter 2

Materials And Method

This chapter describes the experimental apparatus used in this work. First, a brief overview of the implementation of the Magneto-Optical Trap (MOT) for rubidium-87 atoms (^{87}Rb) is presented (this setup was already available prior to the present thesis work). The chapter then focuses on the work carried out during this thesis, namely the implementation of the Optical Dipole Trap setup employed to transfer and trap atoms from the MOT.

Although beyond the scope of the current work, it is worth noting that the long-term objective of this experiment is to confine the atoms within the core of a hollow-core fiber.

2.1 Magneto-optical trap

2.1.1 The Vacuum Chamber

2.1.2 The laser system

2.1.2.1 Master laser

2.1.2.2 Cooler and repumper lasers

2.2 Optical dipole trap

This section describes the implementation of the Optical Dipole Trap (ODT) setup used to transfer atoms from the MOT into a conservative optical potential. The ODT beam is guided through a Hollow-Core Photonic Crystal Fiber (HCPCF). As mentioned in Section 2.1, the HCPCF integrated into the experimental apparatus was already present prior to this thesis work. However, a test HCPCF was employed during the development of the setup in order to optimize the alignment procedure.
During the implementation of the system, dedicated diagnostic tools were also developed to investigate the near-field beam profile and improve the fiber-coupling alignment. The following sections describe the ODT optical configuration and the experimental procedures adopted for the alignment and loading of the trap.

2.2.1 1064nm laser characterization

The ODT is realized by using an IPG Photonics YLR-20-1064-LP-SF laser, a linearly polarized Ytterbium fiber laser that emits 1064 nm light, at a maximum nominal power of 20 W. For this laser model, the mode field diameter changes significantly upon changes of the nominal power set, therefore, the mode profile has been characterized as a function of the power set in order to allow a correct realization of the HCPCF injection setup. This calibration also provides a useful reference for future optimization of the HCPCF injection.

2.2.1.1 Methodology

To characterize the beam properties, its spatial profile is modeled as a Gaussian beam. Therefore, the beam intensity distribution can be described by [5]:

$$I(r,z) = I_0 \left(\frac{w_0}{w(z)}\right)^2 \exp\left(\frac{-2r^2}{w(z)^2}\right), \tag{2.1}$$

where

- z is the axial distance from the beam waist;
- r is the radial distance from the center of the beam;
- $w(z)$ is the beam radius, defined as the radius at which the intensity decreases of a factor $1/e^2$ with respect to the intensity at the beam center;
- $w_0 = w(0)$ is the beam waist, i.e. the beam radius at its focus;
- I_0 is the intensity at the center of the beam at its waist.

The transverse intensity profile follows a Gaussian shape, hence the name of Gaussian beam.

At any axial position z, the beam radius is given by [5]:

$$w(z) = w_0 \sqrt{1 + \left(M^2 \frac{z}{z_R}\right)^2} \tag{2.2}$$

where

$$z_R = \frac{\pi w_0^2}{\lambda} \tag{2.3}$$

is the Rayleigh range, which represent the characteristic distance around the waist over which the beam remains approximately collimated, and $M^2 \geq 1$ is the beam quality factor, accounting for deviations from an ideal Gaussian beam. By acquiring intensity images at multiple longitudinal positions and extracting the corresponding beam radii, Equation (2.2) can be fitted to determine both the waist radius and the waist position.

The experimental setup used for the intensity measurements corresponds to the initial section of the ODT injection setup (see Section 2.2.2.1). With reference to Figure 2.2, measurements were performed on the beam exiting the reflected branch of the second polarization beam splitter (PBS2). The power was adjusted through the regulation of the half-wave plates HWP2 and HWP3, together with the addition of optical density filters when needed. A Complementary Metal-Oxide-Semiconductor (CMOS) camera was employed to take 17 intensity images separated by 25 mm along the propagation axis for the nominal laser powers of 4 W, 7 W, 10 W, 13 W, 16 W and 19 W.

The straightforward approach of extracting the beam radius from each image consists of fitting a 2D asymmetric Gaussian profile and extract the radius along the x and y axis as

$$w_x(z) = 2\sigma_x, \qquad w_y(z) = 2\sigma_y. \tag{2.4}$$

However, fitting equation (2.2) using these extracted radii led to values of $M^2 < 1$ for all the power settings except 4 W, which is unphysical. However, enforcing the constraint $M^2 \geq 1$ resulted in visibly poor agreement between the fitted curves and the measured data. A likely explanation is that the presence of several optical components before the camera introduce aberrations or beam clipping that make equation (2.2) incorrect.

For this reason a second method called $D4\sigma$ was tested [6]. In this method, the beam radius is determined from the second moment of the beam intensity distribution with respect to the intensity centroid. For the x-axis:

$$\hat{\sigma}_x^2 = \frac{\int_{-\infty}^{\infty} (x - x_0)^2 I(x,y)\, dx\, dy}{\int_{-\infty}^{\infty} I(x,y)\, dx\, dy} \tag{2.5}$$

where x_0 is the intensity centroid x-coordinate, and $I(x,y)$ is the beam intensity profile.

Equation (2.2) is then fitted using the radii

$$w_x = 2\hat{\sigma}_x, \qquad w_y = 2\hat{\sigma}_y\,. \tag{2.6}$$

The $D4\sigma$ method should be more robust than the Gaussian fitting when dealing with non-ideal beam profiles, since it accounts for intensity contributions in the beam tails that may not be properly captured by a Gaussian fit. However, the $D4\sigma$ is very sensitive to anisotropic background noise, which can bias both the centroid and second-moment calculations. In the present setup, the presence of the PBS1 and the beam dump that dissipates a large quantity of optical power, generates a non-uniform scattering background. This asymmetry introduces systematic distortions in the $D4\sigma$ analysis. Consequently, even this method did not consistently yield $M^2 \geq 1$ without constraints, although it performed better than the Gaussian fit, giving naturally $M^2 \geq 1$ for 4 W, 16 W and 19 W.

For both analysis methods, a preliminary Gaussian fit was first performed to find approximately the beam center position. A region of interest (ROI) extending to 5σ around the beam center was then selected. Subsequently, either a refined Gaussian fit was performed or the centroid and second moments were calculated to determine the beam radius before fitting Equation (2.2).

2.2.1.2 Results

The beam waist radii and waist positions obtained with the two analysis methods are summarized in Tables 2.1 and 2.2. The fit of Equation (2.2) was performed using the Trust Region Reflective (TRF) algorithm while imposing the constraint $M^2 \geq 1$. The final reported values were obtained by averaging the results of the Gaussian fit and $D4\sigma$ methods. Since the uncertainties returned by the fits were significantly smaller than the discrepancy between the two methods, the uncertainty associated with the averaged value was conservatively estimated as half of the range between the two measurements (the semi-range).

The waist radius as a function of the nominal laser power is plotted in Figure 2.1 top. The results reveal a small but systematic astigmatism, with the waist measured along the x direction consistently exceeding that along the y direction by approximately 1–2%. Furthermore, the graph shows that the waist radius decreases approximately linearly from 7 W to 19 W, indicating a significant power dependence of the laser mode size.

The waist longitudinal position as a function of the nominal laser power is plotted in Figure 2.1 bottom. These values should be interpreted with caution. Imposing $M^2 \geq 1$ allow the fits to correctly match the far points which depends asymptotically on solely w_0 (limit $z \gg 1$ in equation 2.2), but strongly miss the prediction on the point at short distance that give the information on the waist position. Although the extracted values carry significant systematic uncertainty, the observed trend appears physically plausible, with the beam waist shifting downstream as the nominal laser power increases.

Table 2.1: *Comparison of the beam waist w_0 obtained via the D4σ method and the Gaussian fit method. The final values correspond to the average of the two independent estimates. Uncertainties for the individual methods correspond to the fit errors, whereas the uncertainty of the average is given by the semi-range.*

Nominal Power (W)	Estimated waist w_0 (μm)					
	*D*4σ-x	*D*4σ-y	Gaus-x	Gaus-y	Avg-x	Avg-y
4	481 ± 5	474 ± 7	469 ± 4	460 ± 7	475 ± 7	467 ± 8
7	454 ± 3	452 ± 4	419 ± 3	423 ± 5	433 ± 20	430 ± 20
10	403 ± 2	394 ± 3	370 ± 2	364 ± 3	384 ± 20	375 ± 20
13	351 ± 2	343 ± 3	323 ± 4	321 ± 4	334 ± 20	327 ± 20
16	307 ± 2	298 ± 4	283 ± 6	282 ± 6	291 ± 20	284 ± 20
19	288 ± 3	284 ± 4	266 ± 7	268 ± 6	272 ± 20	269 ± 20

Table 2.2: *Comparison of the beam waist position z_0 extracted via the D4σ method and the Gaussian fit method. The final values correspond to the average of the two independent estimates. Uncertainties for the individual methods correspond to the fit errors, whereas the uncertainty of the average is given by the semi-range.*

Nominal Power (W)	Estimated waist position z_0 (mm)					
	*D*4σ-x	*D*4σ-y	Gaus-x	Gaus-y	Avg-x	Avg-y
4	-90 ± 40	-80 ± 50	-40 ± 20	-60 ± 40	-70 ± 20	-70 ± 10
7	20 ± 30	-3 ± 40	-20 ± 20	-40 ± 20	3 ± 20	-20 ± 20
10	90 ± 20	70 ± 30	50 ± 10	40 ± 10	70 ± 20	50 ± 20
13	110 ± 10	90 ± 20	90 ± 20	70 ± 20	100 ± 10	80 ± 20
16	72 ± 9	50 ± 20	100 ± 20	80 ± 20	90 ± 10	70 ± 20
19	70 ± 10	40 ± 10	90 ± 20	70 ± 20	80 ± 10	60 ± 20

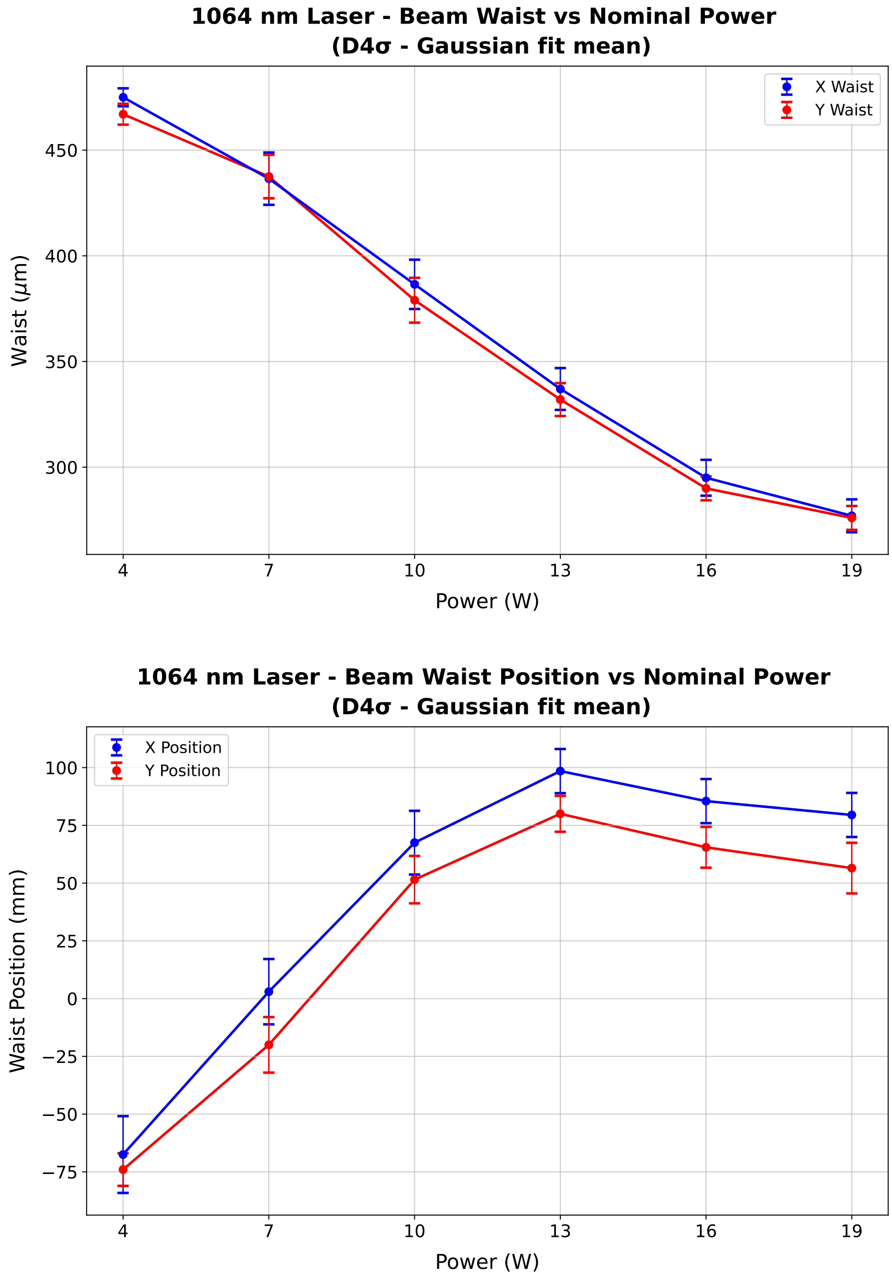

Figure 2.1: *Characterization of the 1064 nm laser beam parameters as a function of nominal output power. (Top) Transverse beam waist size versus power. (Bottom) Longitudinal position of the waist. The reported values correspond to the average of the results obtained using the D4σ and Gaussian-fit methods. Blue and red markers denote measurements along the orthogonal x and y transverse axes, respectively. Error bars indicate the semi-range error of the two analysis techniques.*

2.2.2 Optical setup

In this experiment the ODT light is guided through a HCPCF, which introduces a critical complication in the injection procedure. Indeed, in contrast with standard telecom fibers, HCPCFs are multi-mode fibers. As a consequence, even small deviations of the injected beam from the ideal injection angle excites higher order modes in addition to the fundamental one. This results in distorted output beam profiles, which are not well suited for the realization of an optical conveyor belt based on the superposition of the beam guided by the HCPCF and a beam guided through a conventional telecom fiber supporting only the fundamental LP_{01} mode.
For this reason, particular cure was devoted to the alignment procedure. A significant improvement in the quality of the transmitted mode was achieved by optimizing the injection while inspecting the near-field of the transmitted beam, rather than simply maximizing the transmission efficiency. The following sections describe the injection setup and the horizontal microscope built for the near-field inspection.

2.2.2.1 Injection setup

Figure 2.2 shows the optical setup scheme used for the injection of the laser light in the HCPCF. The scheme is composed of the following elements:

- **Half Wave Plate 1** (HWP1) and **Optical Isolator**. The optical isolator prevents back reflections from propagating towards the laser cavity. High-power lasers are particularly sensitive to back reflections, which can introduce power instabilities and potentially damage the laser system. Since optical isolators operate correctly only for a specific input polarization, HWP1 is used to optimize the polarization of the incoming beam and maximize the transmission through the isolator.

- **HWP2** and **Polarization Beam Splitter 1** (PBS1). These components are used to regulate the optical power delivered to the subsequent stages of the setup. The unwanted fraction of the beam is directed onto a beam dump. This configuration is particularly useful during alignment procedures, where the optical power is reduced to a few milliwatts for safety reasons.

- **HWP3** and **PBS2**. PBS2 splits the beam into two different branches. The first is employed for the HCPCF injection, while the second will be injected into a PANDA polarization-maintaining fiber, which eventually will provide the second beam entering the chamber from above for the realization of the optical conveyor belt. HWP3 is adjusted to have approximately 50/50 power splitting ratio, although this value can be modified if required in future stages of the experiment.

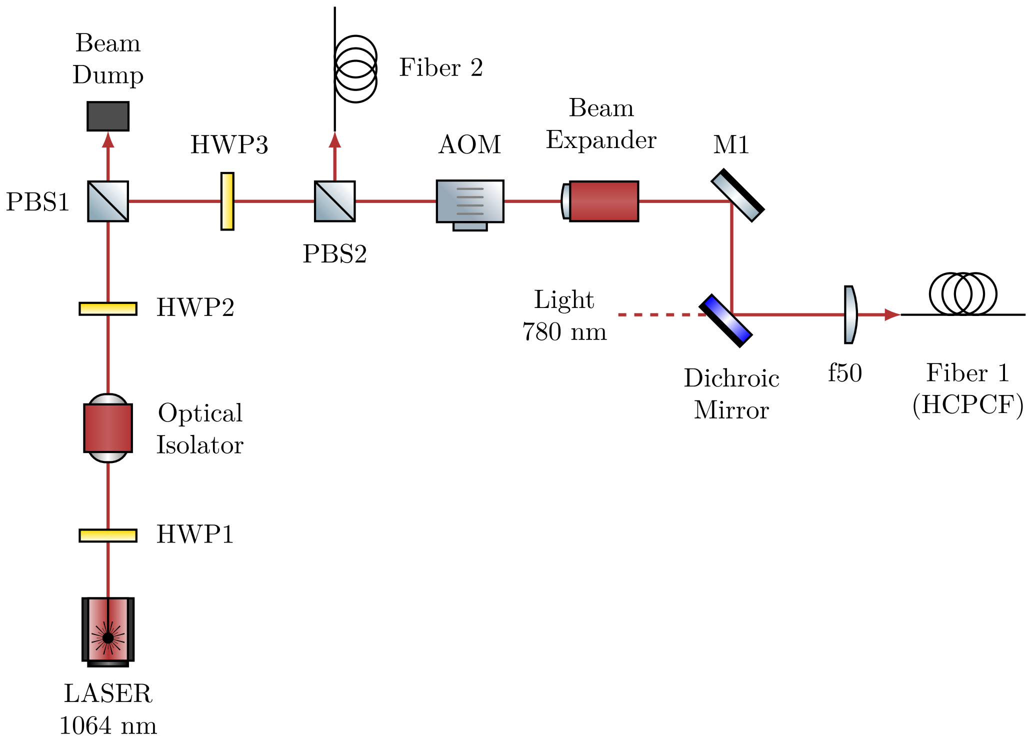

Figure 2.2: *Scheme of the optical setup for 1064 nm ODT beam injection into the HCPCF. The Acousto-Optic Modulator (AOM) operates at 110 MHz, and the subsequent setup utilizes the fist diffraction order. The zero-th order is blocked by a beam dump (not drawn). The 780 nm light setup is not drawn, its presence will be important in the future stages of the experiment for the characterization of the atoms inside the HCPCF core. Fiber 2 injection setup is not drawn as well.*

- **Acousto-Optic Modulator** (AOM). The AOM is employed as a fast switch for the ODT beam. The device operates using an acoustic wave at 110 MHz, generated by a Direct Digital Synthesizer (DDS), causing the first diffraction order to be frequency shifted by the same amount. Although this shift is negligible with respect to the trap depth of the ODT, it will become relevant in the future implementation of the optical conveyor belt.

- **Beam expander**. A ~3X beam expander is employed to increase the beam diameter and collimate it before the focusing conditions for coupling into the fiber.

- **Mirror 1** (M1) and **Dichroic Mirror**. Both mirrors are mounted on tilt stages and their angular adjustment is used for optimizing the fiber injection alignment. The dichroic mirror also combines the 1064 nm ODT beam with a 780 nm beam that will be used in the future for the characterization of the atoms trapped inside the HCPCF core.

- **Focusing Lens** (f50). A lens with focal length $f = 50$ mm focuses the beam into the HCPCF. It is mounted on a horizontal translation stage, providing an additional degree of freedom for optimizing the injection alignment.

- **Fiber 1** (HCPCF). The HCPCF is mounted on an x-y translation stage and terminated with an FC/APC connector. The translation stage allows transverse displacement of the fiber, providing additional alignment degrees of freedom during the injection procedure, although these degrees of freedom were found to have limited impact on the final optimization.

Using the beam waist measurements in Section 2.2.1.2, a nominal laser power of 10 W was selected as the optimal operating point. Since the fitted waist positions were found to be affected by large systematic issues, the beam waist was assumed to be located at the output face of PBS2. The radius at the entrance of the beam expander can be estimated using (2.2), and the measured distance of approximately 175 mm between PBS2 and the beam expander (see figure 2.4). Assuming an ideal beam $M^2 = 1$, using the x waist value for 10 W of $w_0 = 384\ \mu$m, the beam radius at the entrance of the beam expander is estimated to be w(beam expander entrance) $\simeq 414\ \mu$m.
The beam expander provides a magnification factor of three, yielding w(beam expander exit) $\simeq$ 1240 μm. At this stage the corresponding Rayleigh range, calculated using 2.3 is $z_R \simeq$ 4.5 m. Since the remaining propagation distance before the injection optics is negligible compared to the Rayleigh range, the beam considered approximately collimated. Under this assumption, the beam waist after the injection lens can be estimated using [5]:

$$w_{\text{out}} = \frac{\lambda f}{\pi w_{\text{in}}}, \tag{2.7}$$

where $f = 50$ mm is the focal length of the injection lens. Propagating the uncertainty, one gets:

$$w_{\text{inj}} \simeq (13.6 \pm 0.6)\ \mu\text{m}.$$

This result is obtained in the ideal case, the real waist at the injection is expected to be closer to the target value of approximately $\sim 18\ \mu$m. For instance, assuming a more realistic beam quality factor of $M^2 = 1.2$ leads to

$$\tilde{w}_{\text{inj}} \simeq (15.1 \pm 0.5)\ \mu\text{m}$$

Additional deviations from the ideal prediction are expected due to the AOM, which distorts the beam profile by compressing it along the y direction and expanding it along the x direction.
To verify these predictions, the beam profile after the injection lens was characterized following a procedure analogous to that described in Section 2.2.1, yielding

$$w_{\text{inj-}x} = (17.1 \pm 0.7\ \mu\text{m}) \qquad w_{\text{inj-}y} = (19.3 \pm 0.4\ \mu\text{m}),$$

sufficiently close to the target waist.

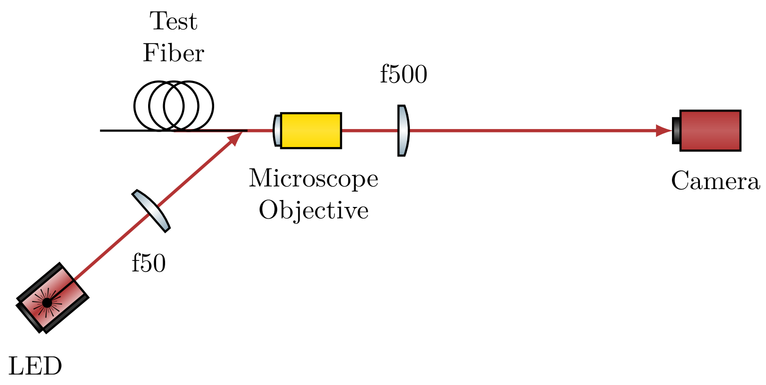

Figure 2.3: *Scheme of the optical setup for the horizontal microscope.*

2.2.2.2 Horizontal microscope

Initially, the injection into the HCPCF installed inside the vacuum chamber was attempted by using a Complementary Metal-Oxide-Semiconductor (CMOS) camera to monitor the beam profile. However, despite the effort, strongly deformed mode profiles were obtained.

In order to better investigate the injection procedure and simplify the alignment process, a separate test HCPCF was employed outside the vacuum chamber. The test fiber consisted of an approximately 50 cm HCPCF segment with the same FC/APC connector as the fiber in the chamber. The test fiber allowed an easy measurement of the efficiency on the exit facet of the beam without any window in the path, and being outside the vacuum chamber allow for easier mode imaging. Moreover, since the two fibers shared the same connector, the optimal alignment conditions identified with the test fiber were expected to provide a good starting point for the alignment of the HCPCF installed inside the chamber. Nevertheless, even with the test fiber, optimization based solely on transmission efficiency and far-field inspection still resulted in poor beam quality.

The fiber manufacturer suggested that the optimal injection condition should be determined by inspecting the shape of the near-field of the transmitted mode. For this purpose, a horizontal microscope system was developed to inspect the HCPCF facet and to investigate the near-field profile of the transmitted mode. The optical scheme of the microscope is shown in Figure 2.3.

The fiber was positioned inside a 90° V-groove of depth 350 μm. The fiber was fixed inside the groove by means of a thin foam layer attached to a tiltable mount using adhesive tape. The coating was removed from the fiber end, leaving approximately 15 mm of suspended fiber in order to allow lateral illumination of the facet.

The HCPCF facet was illuminated by a red LED (wavelength $\lambda = 635$ nm) focused by an $f = 50$ mm lens onto the final suspended portion of the fiber. Part of the red light propagated through the silica structure forming the kagome lattice, making them visible on the microscope camera. The observation of the kagome structure was used to optimize the microscope focus and to inspect the integrity and cleanliness of the fiber facet.

The light emerging from the fiber was collected by an optical microscope objective lens (Olympus PLN10X objective, nominal focal length $f = 18$ mm, working distance $d = 10.6$ mm) and subsequently re-imaged onto a CMOS camera by means of an $f = 500$ mm lens. The expected magnification is $M = f_{\mathrm{tube}}/f_{\mathrm{lens}} \approx 27.8$. The objective is mounted on a horizontal translation stage used for the focussing. The camera was connected to a computer for real-time imaging of the transmitted near-field mode. In addition, a powermeter could be inserted between the fiber output and the objective lens in order to evaluate the coupling efficiency by comparing the transmitted power with the optical power measured before the injection stage. Finally, a black paper tube was placed between the imaging $f = 500$ mm lens and the camera to reduce background light from the LED illumination and from ambient room light.

The alignment procedure was carried out by optimizing the transmitted near-field profile in order to obtain a mode as close as possible to a single-lobed structure, results and further details are found in Section 2.2.3.

Figure 2.4 shows the top-down photographs of the complete experimental apparatus, including both the injection setup and the horizontal microscope system, and figure 2.5 shows a detailed view of the horizontal microscope.

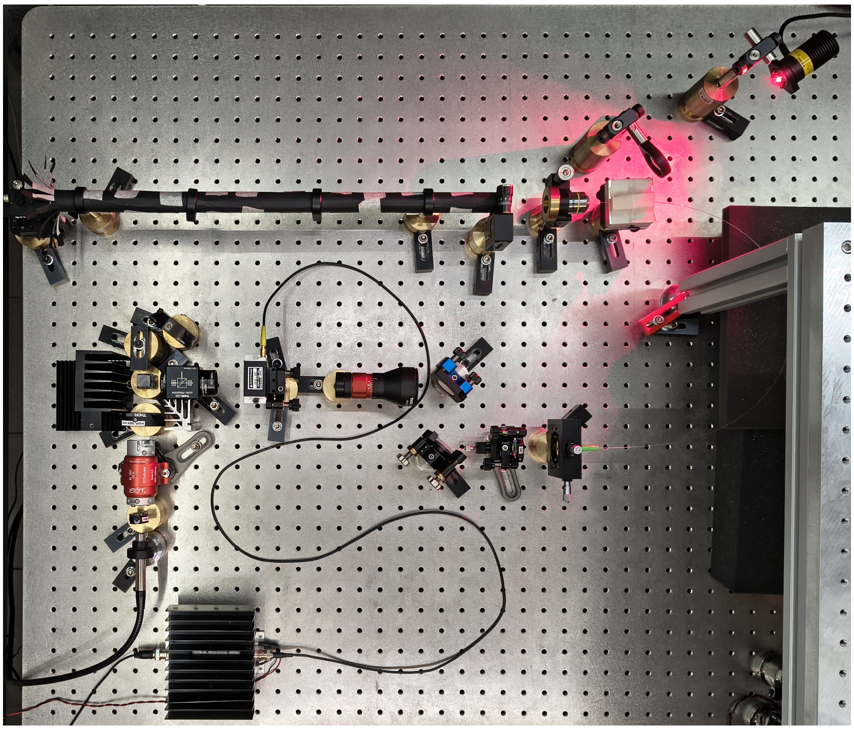

Figure 2.4: *Top-down photograph of the complete experimental apparatus. The horizontal microscope, shielded by the black paper tube, is visible at the top of the image. The optical injection setup is located in the center, while the RF amplifier driving the AOM is placed at the bottom. For reference, the distance between the screw holes on the breadboard is 2.5 cm.*

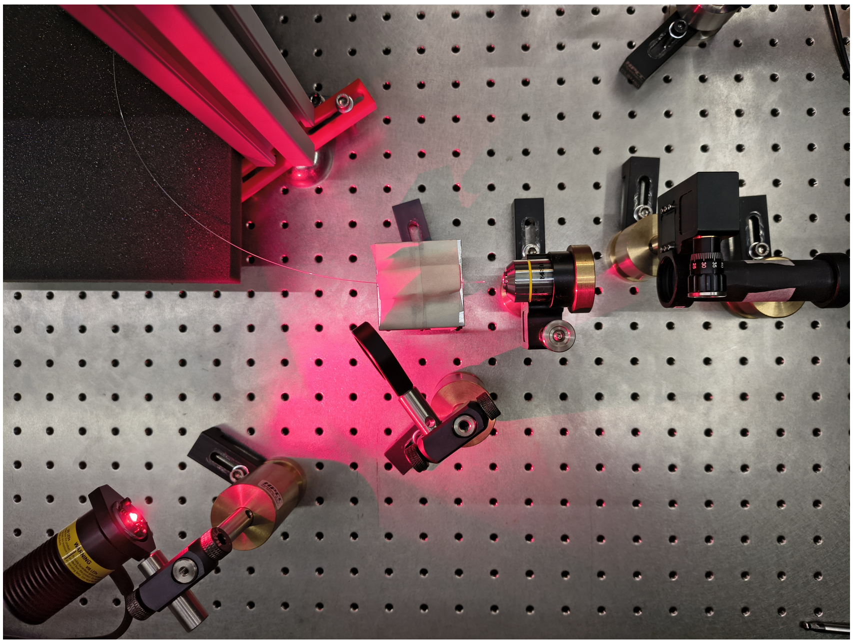

Figure 2.5: *Detailed top-down view of the horizontal microscope. The setup highlights the test fiber, the red LED providing lateral illumination to the fiber facet, the microscope objective lens and the refocusing lens.*

2.2.2.3 Chamber fiber optical imaging setup

After the optimization and the validation of the injection alignment procedure using the test fiber, the same approach was applied to the fiber inside the vacuum chamber.

To investigate the near-field profile, a dedicated optical imaging setup was build on a breadboard positioned above the vacuum chamber (Figure 2.6). Since the optical fiber is located inside the chamber, a first stage microscope was designed to create a virtual image of the fiber facet outside the chamber. For this purpose a f150 lens (f = 150 mm) was placed approximately 150 mm above the fiber facet, directly above the chamber window, such that the light emerging from the fiber is collimated. The collimated beam propagate through an opening and is redirected horizontally by a mirror (M1) tilted at 45°. A second lens f300 (f = 300 mm) refocuses the beam, forming an inverted image of the fiber facet with a magnification of $\times 2$.

A second imaging stage, consisting of lenses L3 and L4, acted as a microscope for this intermediate image and projected a magnified image onto the sensor of a camera. The focal lengths of L3 and L4 were modified several times throughout the experiment to optimize the field of view and magnification. The configuration used most frequently employed an L3 lens with f = 100 mm and an L4 lens with f = 500 mm, resulting in an additional magnification of $\times 5$. Combined with the first imaging stage, this yielded an overall magnification of approximately $\times 10$, which was used for the majority of the measurements.

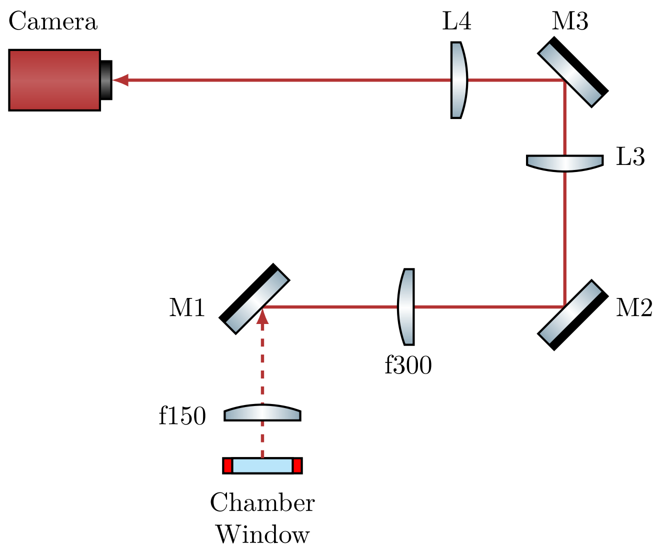

Figure 2.6: *Scheme of the optical setup used to image the fiber inside the vacuum chamber. The optical path between the chamber window and the mirror M1 is vertical (characterized by the dashed beam line), whereas the rest of the setup is arranged horizontally on a breadboard mounted above the chamber. The focal length of lenses L3 and L4 were varied several times during the experiment. Most images where acquired using an* $f = 100$ mm *and an* $f = 500$ mm *providing an overall magnification of approximately* $\times 10$ *combined with the first imaging stage.*

2.2.3 Injection alignment and near-field imaging

For a conventional single-mode telecom fiber, the alignment procedure is straightforward: it is sufficient to inject a low-power beam of few milliwatts, monitor the transmitted power at the fiber exit, and adjust the injected beam to maximize the guided optical power.

By contrast, aligning the injection of a multimodal HCPCF is significantly more complex. The silica jacket that surrounds the kagome core can guide the light with coupling efficiencies as high as 60%, although with badly shaped mode (see figure 2.7). Considering also that the jacket is large compared to the core, monitoring just the coupling efficiency is particularly ineffective, as the optimization can easily get trapped in a local maximum associated with jacket guiding.

As suggested by the manufacturer, the best procedure for the injection alignment resulted to be monitoring the near-field mode shape at the fiber exit. The complete alignment procedure begins with the use of a powermeter to ensure the impinging beam hits the fiber, even on the jacket. Then switch to live monitoring of the near-field with a camera and optimize its shape by tuning the degrees of freedom of the injection setup. Coupling efficiency is measured only after the mode profile has been optimized.

Applying this protocol, the test fiber injection was optimized employing the injection optical setup and the horizontal microscope described in Section 2.2.2. This approach yielded to a good result and thus was subsequently applied to align the injection into the

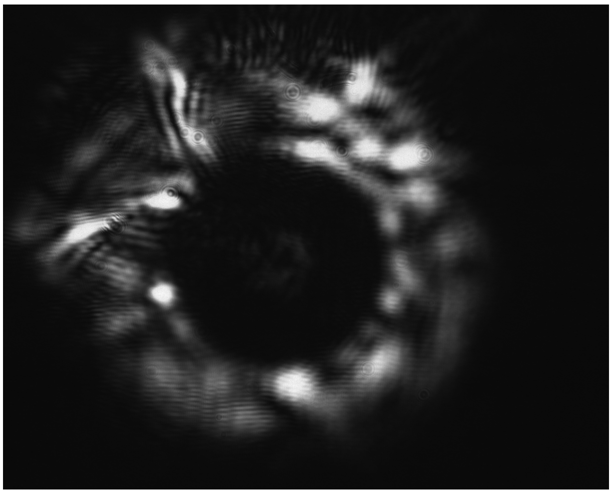

Figure 2.7: *Example of the near-field of a mode guided by the test fiber jacket. The light is mainly guided by the silica jacket surrounding the kagome structure, while only a weak signal is visible in the core. Note that this strongly deformed mode had a coupling efficiency of* $\sim 60\%$.

fiber in the vacuum chamber.

2.2.3.1 Test fiber

The test fiber was cleaved prior to inspection. Since the horizontal microscope has a sufficient resolution to resolve the kagome structure of the fiber, it was exploited for the focusing procedure. By monitoring the camera imaging in real-time, the kagome structure was focussed using the translation stage of the microscope objective. The objective refractive index can be considered roughly constant across the visible spectrum and into the near-infrared (NIR). This guarantees that when the structure (illuminated by red light) is in focus, the guided mode (1064 nm NIR light) is also in focus. Figure 2.8 shows an image of solely the kagome structure alongside an image of the kagome structure together with some guided light. The image show that the kagome structure is intact confirming that the cleaving was correctly performed. Furthermore, the mode is clearly guided by the core and exhibits a single lobe, however, the shape is not symmetric with respect to the fiber axis, indicating the excitation of higher order LP_{11} mode alongside the fundamental LP_{01}.

The mode seen in Figure 2.8 right was associated with a transmission efficiency of 63%. By optimizing the alignment using the horizontal microscope the mode shape was optimized, and an efficiency of 80% has been achieved. For this optimization two screw of the mirror mounts were adjusted at the same time as a continuous walk-off, therefore performing an optimization remaining as long as possible in a good injection condition. The resulting mode is shown in Figure 2.9 left. Figure 2.9 right displays another mode obtained with nearly 80% of injection efficiency, where the surrounding kagome structure remains visible.

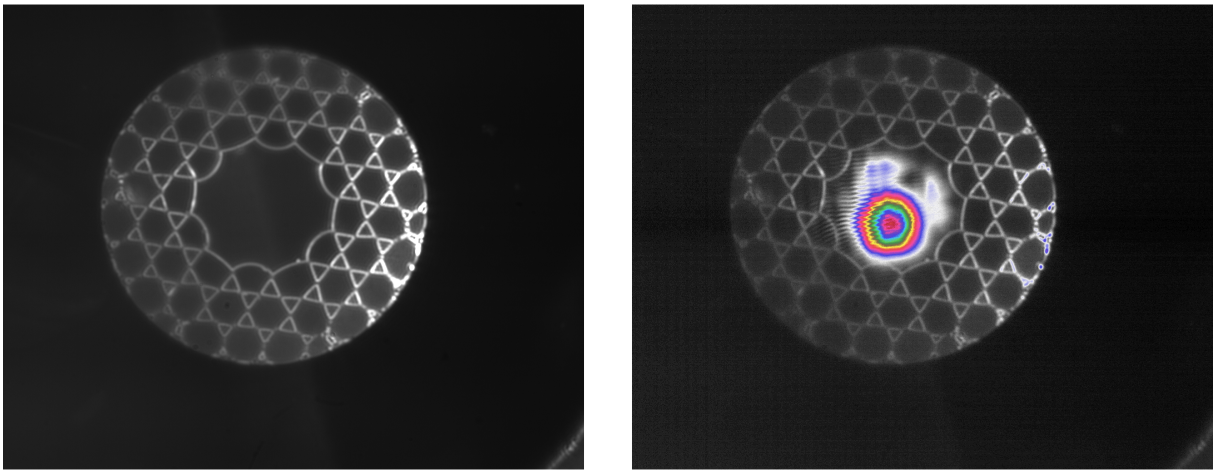

Figure 2.8: *Images taken with a preliminary horizontal microscope setup of the HCPCF kagome structure (left) and of the transmitted mode together with the kagome structure, with a pseudo color map (right). These images were taken with a preliminary setup of the horizontal microscope that provided a more homogeneous illumination, with the red light incident from two direction, making the kagome lattice distinctly visible.*

A final optimization was performed without the ocular lens by translating the objective slightly past the working distance and placing the camera approximately 2 m away. With this configuration, the magnification achieved is about $\times 70$ (more than doubling the magnification of the horizontal microscope). At this distance, the light scattered from the kagome structure becomes very faint, complicating the focusing procedure as its signal approaches the noise level. However, employing the FLIR camera, which has a lower read noise and a high sensitivity, the kagome lattice remained resolvable, as shown in Figure 2.10. This figure shows clearly that the near-field mode assumes a hexagonal shape, perfectly fitting the core: as expected the field inherit the symmetry of the medium in which it is confined.

The advantage of this extended projection distance is the significant expansion of the mode, the pixel grid interference patterns disturb the shape far less, allowing finer spatial details to be resolved, further improving the alignment procedure. With this method an efficiency of 94% was achieved. The resulting mode (Figure 2.11 left) is still slightly asymmetric, indicating a residual superposition of higher order modes with the fundamental one.

Finally, a sweep of laser nominal power was conducted, which changes also the beam waist at the fiber injection. At every tested power the injection alignment was re-optimized, and the best mode profile was obtained at 6.5 W, with an efficiency of 96%. The mode is shown in Figure 2.11 right, in this case the shape is far more symmetric than the previous iterations, which, together with the high transmission efficiency, suggests a significant high share of the fundamental LP_{01} mode.

Qua manca qualcosa sulle due fotcamere, e anche un riferimento alla HCPCF (aggiungerò una sezione)

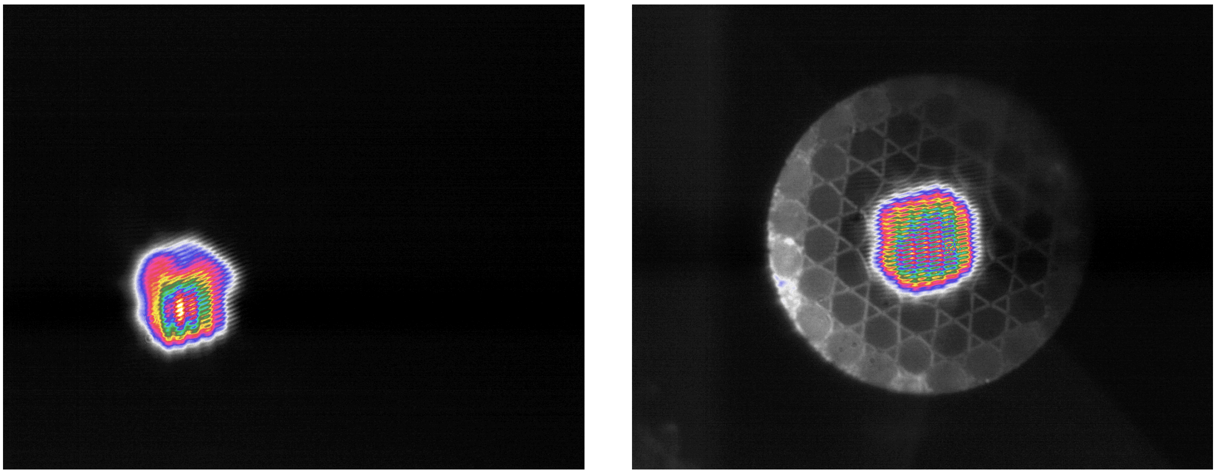

Figure 2.9: *Images captured with the final horizontal microscope setup. A mode with* $\sim$ 80% *of transmission efficiency (left) and a guided mode with the kagome structure visible (right).*

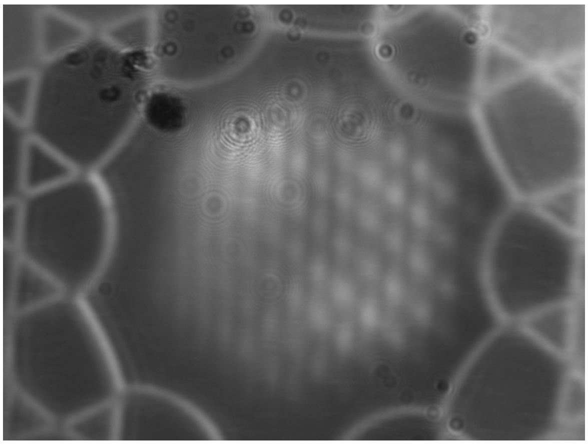

Figure 2.10: *Images obtained with the FLIR camera using the microscope objective without the ocular lens. The mode clearly exhibits a hexagonal shape. Thanks to the FLIR high sensitivity and low read noise, the kagome structure remains visible even in this configuration. Artifacts in the top-left corner are dust grain deposited on the camera sensor.*

2.2.3.2 Chamber fiber

After validating the effectiveness of the alignment procedure using the test fiber, the same method was applied to the fiber installed inside the vacuum chamber. Unfortunately, it was not possible to properly illuminate the fiber, and the kagome structure could not be visualized. Using the same 635 nm LED employed for the test fiber, several configurations of focusing lenses and LED positions were tested, however, no light was successfully coupled into the kagome structures. In addition, due to physical constraints imposed by the vacuum chamber geometry, it was not possible to directly illuminate the fiber facet from the outside. One possible limitation might be the presence of a translucent plastic casing surrounding the fiber. The casing may diffuse the light of the LED preventing it to reach the kagome structure. Unlike the images obtained with the test fiber, in this case

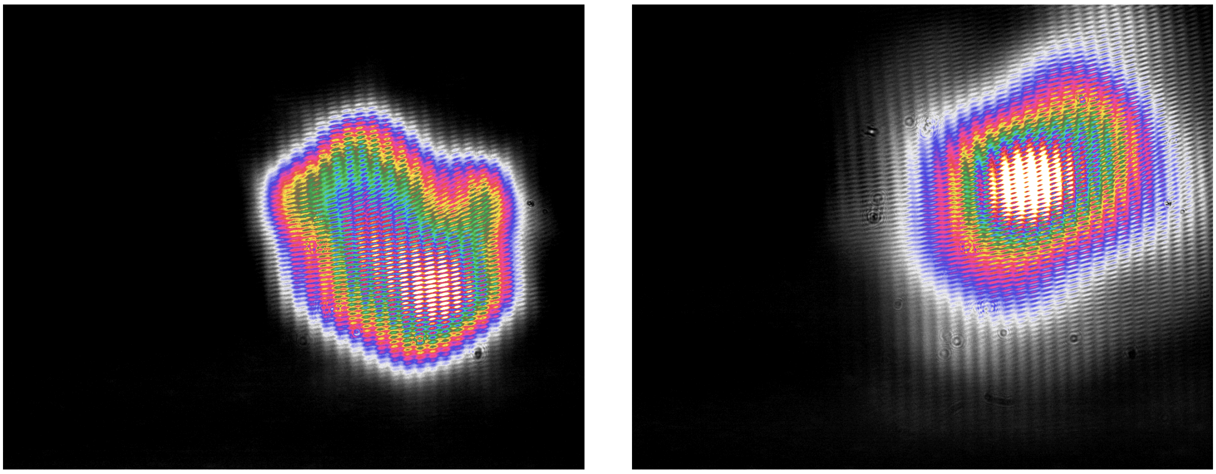

Figure 2.11: *Images obtained with the objective without the ocular lens. Mode with* 94% *of injection efficiency (left). Optimized mode with nominal laser power set to 6.5 W which achieved* 96% *of transmission efficiency (right).*

the jacket appears illuminated, suggesting that the light propagated in the jacket rather than reaching the fiber kagome structure.

Furthermore, the need to create a virtual image outside the chamber required the use of multiple lenses (see Section 2.2.2.3). While this approach enabled imaging of the fiber facet, it also introduced non-negligible chromatic aberrations. Figures 2.12 left and 2.12 right show respectively a preliminary image of the focused fiber and the same configuration with light guided through the fiber. In the latter case, the optical mode is in focus while the fiber facet appears out of focus.

The mode obtained after the injection alignment procedure is shown in Figure 2.13, which exhibited $\sim 50\%$ of transmission efficiency. It should be noted that this efficiency values was measured after the light propagated through a chamber window, two lenses and two mirrors, all of which introduce transmission and reflection losses. Moreover, the fiber installed in the chamber is longer than the test fiber. Although propagation losses associated with the additional length are should be negligible, the increased length requires several bends in the fiber routing, which are known to induce leakage losses in optical fibers.

The best mode obtained after the optimization procedure is displayed in Figure 2.13. It shows a clear asymmetry, indicating the presence of a superposition of various LP modes. Although, it should also be considered that, in order to enter the vacuum chamber, the fiber must be tightened in a ferrule by a swage-lock fitting. This mechanical stress can deform the fiber microstructure, altering its guiding properties and degrading the mode shape quality.

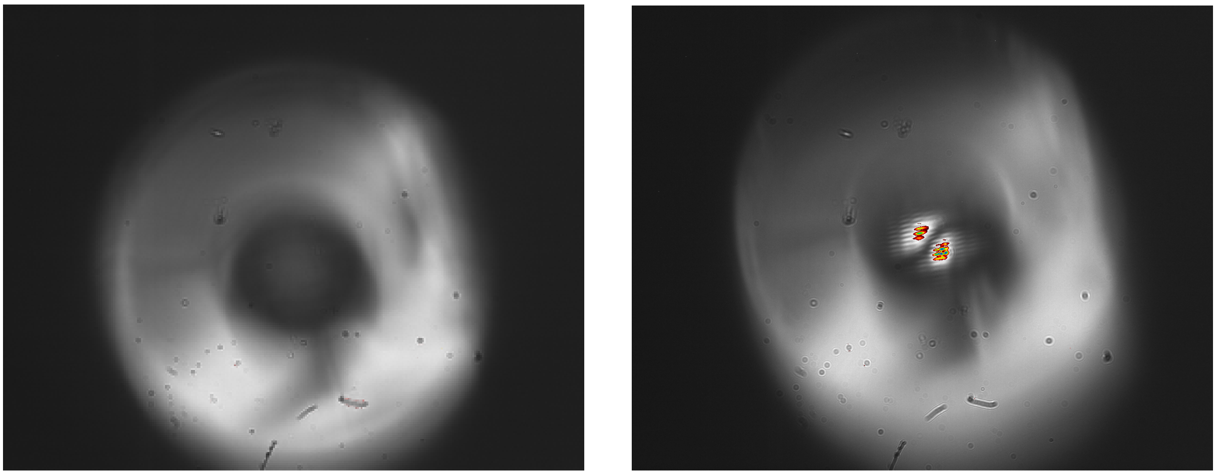

Figure 2.12: *Preliminary images taken with a zoom about ×10 of the fiber facet in the vacuum chamber (left) and the fiber with a non-optimized mode (right). In contrast to the test fiber images, here, the kagome structure is not illuminated, whereas the outer jacket is. The mode exhibits a clear* LP_{11} *profile. Note that focusing the mode bring the fiber out of focus, this makes the mode seem non-centered with respect to the fiber core.*

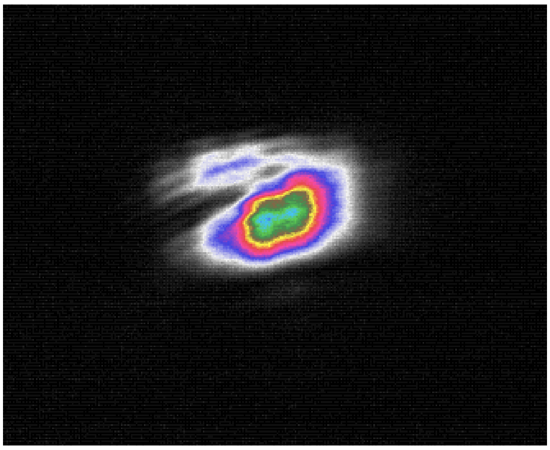

Figure 2.13: *Final mode profile obtained after the injection optimization (magnification about ×10). The mode shape is a bit deformed, meaning that it is a superposition of fundamental LP modes. The image is cropped.*

2.3 Experimental Procedures and analysis methodology

The first step of the experimental sequence is the capture of atoms in the MOT, using the optimal parameters previously determined in [7] for this apparatus. The MOT is realized with a magnetic field gradient of $B' = 9.32$ G/cm (corresponding to a current of 8 A), a cooler laser detuning of $\delta = -13$ MHz, and all compensation coil currents set to 0 mA. The MOT is loaded for a total duration of 5 s.

After the loading phase, the magnetic field is switched off, initiating the optical molasses phase. In the absence of the magnetic field, the atomic temperature decreases, however the confinement force is lost and the atomic cloud begins to expand. At the same time, under the influence of gravity, the cloud starts to fall while experiencing a viscous-like damping force due to its interaction with the laser light. The temperature achieved during the molasses phase scales as $I/|\delta|$, I is the laser intensity and δ is the detuning. However, this relation remains valid as long as $|\delta|$ is sufficiently small to not reduce significantly the optical cycle rate $\Gamma_\mathrm{p} \propto 1/\delta^2$, QUA SISTEMA. Furthermore, optimal molasses performance is obtained when the magnetic field is effectively zero. For this reason 5 ms after the MOT magnetic field is switched off, the shim coil currents are adjusted to compensate for the Earth's magnetic field and any residual magnetic fields present in the experimental environment. The molasses is performed with a detuning different from that used for the MOT phse, it is changed with a ramp of 20 steps with a total duration of 5 ms. The ramp stars 5 μs after the shim coils are changed to their molasses value.

2.3.1 Atom Imaging

Two different CMOS cameras were used to image the atoms inside the vacuum chamber:

- **Teledyne FLIR BFS-U3-04S2M-CS**, monochromatic camera mounting the Sony IMX287 sensor, with a resolution of 720×540, and a pixel size of 6.9 μm $\times$ 6.9 μm.

- **Allied Vision Manta G-419**, monochromatic camera mounting the CMOSIS/ams CMV4000 CMOS sensor, with a resolution of 2048×2048, and a pixel size of 5.5 μm $\times$ 5.5 μm.

The FLIR camera was primarily used during the optimization of the molasses phase, whereas, the Manta camera was employed for measurements of the atom capture efficiency of the ODT. The Manta was preferred with respect to the FLIR because of its higher sensitivity at 780 nm and its higher pixel density, which helped the imaging of the small region where the ODT is present.

Both cameras were equipped with a Thorlabs MVL50M23 adjustable objective lens with a focal length of 50 mm, and a minimum working distance of 200 mm, corresponding to a minimum magnification of ~ 0.33. The objective was adjusted to focus on the HCPCF inside the vacuum chamber, ensuring that the atomic cloud, located at a similar distance, was also in focus. Both cameras were externally triggered by signals sent by the FPGA control system.

Since the temperature determination of the atomic cloud requires knowledge of its physical dimensions (see Section 2.3.2), the FLIR camera had to be calibrated.

The first calibration method was based on the measure of the diameter in pixel of the fiber inside the vacuum chamber. By turning on the MOT beam, the fiber gets illuminated, and a picture is taken. Using this method the fiber diameter of ~ 316 μm was found to correspond to 8 pixels. This yielded a calibration factor of 39.5 μm/px, corresponding to a magnification of approximately 0.17. However, the measurement uncertainty was significant: assuming an uncertainty of one pixel on each edge of the fiber image, the measured diameter becomes (8 ± 2) px, resulting in a calibration factor of (40 ± 10) μm/px. Consequently, this method provides only a rough estimate of the imaging magnification.

A second and more accurate calibration method was therefore employed. A mirror was positioned in front of the camera without modifying the objective setting, and a ruler with a visible scale was positioned so that it was in focus. This ensured that the ruler was located at the same distance from the objective as the fiber inside the vacuum chamber. With this method 2 cm corresponded to (471 ± 2) px. Since the instrumental uncertainty associated with a single measurement was negligible, the calibration was performed by taking five independent measurement, each time the ruler was moved out of focus and repositioned in focus to have five independent. The resulting five calibration factors were averaged, and the standard deviation of the mean was taken as the uncertainty. This procedure yielded a final calibration factor of (43.4 ± 0.3) μm/px, corresponding to a magnification of 0.16.

2.3.2 Temperature measurements

Temperature measurements were performed toe optimize the molasses parameters, with the goal o achieving the lowest possible temperature.

The temperature of the atomic sample was determined by measuring its ballistic expansion during free fall. Assuming a non-interacting gas and considering a single spatial coordinate, if an atom is found at x_0 at $t = 0$, after a free expansion time t its position is given by

$$x(t) = x_0 + v_x t. \tag{2.8}$$

Taking into account the distributions of both positions and velocities, and assuming that these quantities are statistically independent, the variance of their sum is equal to the sum of their variances. Therefore,

$$\sigma_x^2(t) = \sigma_{x,0}^2 + \sigma_{v_x}^2 t^2, \tag{2.9}$$

where $\sigma x, 0$ is the standard deviation of the position distribution at $t = 0$.
According to classical energy equipartition theorem, the temperature associated with a given spatial direction is related to the squared mean velocity along that direction:

$$\frac{1}{2} m \langle v_x^2 \rangle = \frac{1}{2} k_B T_x, \tag{2.10}$$

where m is the atomic mass and k_B is the Boltzmann constant.
In the frame of reference of the cloud center of mass, the atomic cloud is globally at rest, thus $\langle v_x \rangle = 0$ and $\langle v_x^2 \rangle = \sigma_{v_x}^2$.
Since the velocity follow the Maxwell–Boltzmann distribution, it follows that

$$\langle v_x^2 \rangle = \sigma_{v_x}^2 = \frac{k_B T_x}{m} \tag{2.11}$$

Substituting this result into 2.9 one gets:

$$\sigma_x^2(t) = \sigma_{x,0}^2 + \frac{k_B T_x}{m} t^2. \tag{2.12}$$

Using the atomic mass of the ^{87}Rb

$$m = m_A(^{87}R) = 86.9 \text{ u} \approx 1.443 \times 10^{-25} \text{ kg}, \tag{2.13}$$

and acquiring images at different times of flight (TOF), equation (2.12) can be fitted obtaining T_x as a fitting parameter.
The complete acquisition and analysis procedure is summarized below:

1. First a MOT is loaded, followed by the molasses phase.

2. After the molasses stage, all cooling beams are switched off, allowing the atomic cloud to expand and fall freely. Using the FLIR camera images are taken at different TOFs, obtaining the intensity distribution $I(TOF)$. For each TOF a corresponding background image $I_{\text{bkg}}(TOF)$ is also acquired. The background image is taken by waiting a sufficiently long time (1s) ensuring that the cloud has completely disappeared and only the background signal remains.

   It should be noted that, due to the time required by the FLIR camera to transfer an image to the computer, it is not possible to take multiple TOF image during the same experimental run, therefore each image requires a new MOT loading and molasses sequence.

3. The background is subtracted

$$I_{\text{clean}}(TOF) = I(TOF) - I_{\text{bkg}}(TOF), \tag{2.14}$$

and the associated Poisson uncertainty is calculated as

$$\sigma_{\text{N}} = \sqrt{I(TOF) + I_{\text{bkg}}(TOF)}. \tag{2.15}$$

4. A two-dimensional Gaussian is fitted to each image (TRF minimization algorithm) to extract the cloud widths σ_x and σ_y along the two principal axis.

5. Equation (2.12) is fitted independently for both directions, obtaining the temperatures T_x and T_y. Then, The final temperature T is calculated as their average.

2.3.3 ODT loading measurements

Chapter 3

Result And Discussion

3.1 Molasses temperature optimization

3.2 ODT loading optimization

3.2.1 Shifted MOT

3.2.2 Compressed MOT

Bibliography

[1] Rudolf Grimm, Matthias Weidemüller, and Yurii B. Ovchinnikov. *Optical dipole traps for neutral atoms*. 1999. arXiv: physics/9902072 [physics.atom-ph]. URL: https://arxiv.org/abs/physics/9902072.

[2] Daniel A. Steck. *Quantum and Atom Optics*. Revision 0.16.8, accessed: 2026-06-01. 2026. URL: http://steck.us/teaching.

[3] Harold J. Metcalf and Peter Straten. *Laser Cooling and Trapping*. Graduate Texts in Contemporary Physics. Springer New York, NY, 1999. DOI: https://doi.org/10.1007/978-1-4612-1470-0.

[4] Claude Cohen-Tannoudji. "Atomic Motion in Laser Light". In: *Fundamental Systems in Quantum Optics*. Ed. by J. Dalibard, J. M. Raimond, and J. Zinn-Justin. Vol. LIII. Les Houches Summer School. Elsevier Science Publishers, 1992.

[5] Bahaa E. A. Saleh and Malvin Carl Teich. *Fundamentals of Photonics*. 3rd. Hoboken, NJ: John Wiley & Sons, 2019. ISBN: 978-1-119-50687-4.

[6] A. E. Siegman. "How to (Maybe) Measure Laser Beam Quality". In: *DPSS (Diode Pumped Solid State) Lasers: Applications and Issues*. Optica Publishing Group, 1998, MQ1. DOI: 10.1364/DLAI.1998.MQ1. URL: https://opg.optica.org/abstract.cfm?URI=DLAI-1998-MQ1.

[7] Letizia Loparco. "Characterization of a magneto-optical trap of 87 Rb atoms". Supervisor: Prof. Francesco Minardi. MA thesis. University of Bologna, Italy, 2025. URL: https://amslaurea.unibo.it/id/eprint/35347.